OpenPantry

Administrator Handbook

FOR ADMINISTRATORS

Setup, configuration, security, reporting, and day-to-day operation of OpenPantry.

Revised July 2026 • Version 1.0

OVERVIEW

About OpenPantry

OpenPantry is a self-contained PHP + SQLite application that helps a food pantry check items out the door, keep a live inventory, and forecast what to reorder — with no database server, build step, or external dependencies to install. It is a hybrid Just-In-Time inventory system built for smaller pantries with limited storage and high-demand essentials.

Every outbound channel — grocery-style barcode checkout, the Menu Counter ordering kiosk, home deliveries, community events, and OrderAhead imports — feeds one inventory and one demand model, so the reorder report always sees the whole picture. This handbook is your reference for standing the system up, keeping it secure, and running it day to day.

> **WHAT YOU'RE RESPONSIBLE FOR**
> As administrator you own the four things volunteers never touch: the **OpenAI key & settings**, the **network / hours access gate**, the **admin password & encryption key backup**, and the **reorder cron job**. Each has its own section below.

GETTING INSTALLED

Requirements & deployment

- Any web host with **PHP 8+** and the **pdo_sqlite** extension (standard on every default install).
- For field encryption, the PHP **sodium** extension (built in on PHP 7.2+).
- The application folder (and `menucounter/`) must be **writable** by the web server so the databases and encryption key can be created on first hit.

1. Copy the whole OpenPantry folder to your host. The nested Menu Counter app ships inside it — no separate deploy.
2. Browse to the app root. The schema builds itself, the produce table seeds, and `openpantry.db` plus `menucounter/picklist.db` are created automatically.
3. Open **Settings** and complete first-time configuration (next section).

DO THIS BEFORE OPENING

First-time configuration (Settings)

Everything below lives on the **Settings** page:

- **Administrator password** — change it from the default `admin` immediately. It is stored one-way hashed and can't be recovered, only reset.
- **OpenAI API key** — powers automatic brand→generic naming on new barcodes. Use the **test** button to confirm it works. Stored encrypted at rest.

- **Network Access IP** — set to your pantry's public Wi-Fi address so the kiosks only work on-site. Leave blank during setup to avoid locking yourself out, then set it.
- **Allowed hours** — optional weekly schedule that closes the kiosks outside service times.
- **Administrator email** — where reorder-reminder digests are sent.
- **Email notifications (SMTP)** — optional authenticated sending; otherwise the app uses PHP `mail()`. Use **Send Test Email** to verify delivery.
- **Produce tare** — ounces subtracted from hand-typed produce weights (not scale readings).
- **Par-level defaults** — default lead time and safety-stock Z used by the Order Now report.
- **Food pantry name & logo** — branding shown in the header and on printed sheets.

PROTECT THIS ABOVE ALL

Security & data privacy

- **Admin password** is stored as a one-way hash (`password_hash`); even the running app can't read it back. The login cookie is derived from the hash, so changing the password instantly logs everyone out.
- **Network + hours gate** (in `auth.php`) blocks any device that isn't on the allowed IP or is outside allowed hours, with a styled “Access Denied” wall.
- **Field-level encryption** (libsodium) protects sensitive columns at rest: the OpenAI key, the allowed IP, and delivery clients' address / city / phone.

> **TWO TIERS OF ACCESS — AND WHY VOLUNTEERS DON'T LOG IN**
> The **volunteer kiosks are gated by the allowed IP only**: the scanning stations, the Menu Counter order form and pick queue, and the delivery kiosk open straight to the work screen on the pantry network, with **no password**. The **admin password guards the administrative screens** — the dashboard, Settings, Inventory, Restock, Lookup Tables, Reports, the Menu Counter item admin, and the delivery client roster. Setting the allowed IP correctly is therefore what keeps the kiosks safe, so keep it current whenever the pantry's public address changes.

> **BACK UP ENCRYPTION_KEY.PHP**
> The 32-byte key in `encryption_key.php` is generated on first use and is the **only** thing that can decrypt your data. If you lose it, all encrypted fields are gone for good. Back it up, keep it out of version control, and ideally relocate it above the web root via the `OPENPANTRY_KEY_PATH` environment variable or a `FS_ENC_KEY_PATH` constant. On PHP without libsodium, encryption silently degrades to plaintext until a sodium-capable PHP runs.

KEEPING COUNTS HONEST

Managing inventory

- **Inventory page** — the canonical current-count list. Create items here and set each one's unit (`each` or `lb`).
- **Restock** — stage counts and submit a batch that *adds* to inventory, flagged purchased vs. donated (drives the Purchased-% column). No order rows are written, so restocks stay out of usage reports.
- **Deliverable flag** — uncheck it to keep an item in inventory but hide it from the Menu Counter order form.
- **Count-per-case** — set how many units a supplier case holds; the Order Now report then shows a Case Request column (order need ÷ case size, rounded up).
- **Checkout, deliveries, events, OrderAhead** all decrement inventory automatically as orders close or imports run.

BARCODES TO GENERIC NAMES

Lookup tables & naming

OpenPantry stores each item by a plain **generic name** (e.g. “Black Beans”) rather than brand, so demand aggregates cleanly. The first time a UPC is scanned, the app looks it up on Open Food Facts and asks OpenAI to reduce it to a 2–4 word generic, then caches the mapping.

- **Produce** — PLU codes (and 12-digit pantry labels starting with 4) map to produce names; the table ships pre-seeded and you can add more.
- **UPC / Generic Cache** — review and edit AI-derived names in place. If it guessed “Beans” when you wanted “Black Beans”, fix it here and future scans use your edit.

TURNING SCANS INTO DECISIONS

Reports & the demand model

- **Order Now** — the reorder report. Computes a Par Level and how much to order per item.
- **Orders Listing** — every order and its items over a date range (delivery/event orders are tagged).
- **Item Usage** — per-item totals over a date range.
- **Daily Volume** — orders and scans per day.
- **Basket Size** — distribution of items per in-pantry trip over time.

How Par Level is computed

For each item: **Par Level = Forecast(LeadTime) + SafetyStock**, where **SafetyStock = Z × √Variance(LeadTime)** and **Order need = max(0, Par Level – latest inventory count)**. Forecast and variance come from a quasi-Poisson model fitted to each item's weekly scan history, with a time trend and annual seasonality; dispersion inflates the safety stock so it reflects real volatility. Z defaults to 1.65 (~95%). Items with under ~2 months of history fall back to a trailing average and are marked with a `°`.

> **THE FORECAST CACHE IS DISPOSABLE**
> Fits are memoized in the `forecast_cache` table and rebuild on demand. If a report ever looks stale or wrong, that table is safe to delete — it will rebuild from the scan history.

Reorder alerts & email

Set a per-item lead-time alert and it shows as a banner on Order Now when projected days-of-stock drop below the threshold. Tick its **Email** box to also have it emailed. The cron job (`cron_reorder_alerts.php`) mails a digest of triggered, email-flagged items to the administrator address, at most once per ~20 hours per item.

AUTOMATED REMINDERS

The reorder cron job

Add a cron job (cPanel → Cron Jobs) that runs the mailer on your cadence, e.g. daily at 7am:

```
0 7 * * *  /usr/local/bin/php
/home/you/public_html/openpantry/cron_reorder_alerts.php
```

> **CALL PHP BY ITS ABSOLUTE PATH**
> Use the full path to the PHP binary (e.g. `/usr/local/bin/php` on Namecheap-style cPanel hosts). A bare `.php` path can fail silently with “Permission denied” and no email ever goes out. Confirm with **Send Test Email** in Settings first.

THE ORDERING KIOSK

Administering the Menu Counter

- **Item admin** — add, remove, reorder (drag & drop), and toggle items; set category, name, sizes, and the **Family Factor** (multiplied by household size, capped at 5, rounded up) that decides how many units land in the pick queue.
- **Pick queue (Orders)** — the live back-of-house dashboard volunteers pull from; auto-refreshes every 30 seconds.
- **Deduplicate** — merge duplicate item rows that crept in over time.
- **Reports** — item-usage reports with a chart; shoppers are anonymized as “Client N.”
- The Menu Counter shares the same login, allowed-IP, and allowed-hours gate as the rest of OpenPantry, and it hides any item you've marked non-deliverable on the Inventory page.

KEEPING IT HEALTHY

Maintenance & troubleshooting

- **Backups** — back up `openpantry.db`, `menucounter/picklist.db`, and `encryption_key.php` together. The two databases hold your data; the key decrypts it.
- **No email arriving?** Verify the admin email and SMTP in Settings, send a test, and check the cron PHP path (above). Without SMTP the app falls back to PHP `mail()`.
- **New barcode saved with a raw name?** The OpenAI step was skipped (bad/rate-limited key). Fix the key and edit the name under Lookup Tables.
- **“Access Denied” for legitimate staff?** The public Wi-Fi IP changed or you're outside allowed hours. Update the allowed IP in Settings.
- **Report looks off?** Delete the `forecast_cache` table; it rebuilds from scans.

POST THESE AT EACH STATION

Opening checklists (for training)

These are the same checklists in the Volunteer Handbook. Print and post one at each station, and use them when training new volunteers.

CHECKOUT · LASER SCANNER

Open-the-Station Checklist — Scanning Station

Work top to bottom. Tick each box as you go.

- [ ] Power on the Chromebook tablet and connect it to the pantry Wi-Fi (the scanner only works on the pantry network) if it is not already connected.
- [ ] Make sure the handheld barcode scanner is connected to the tablet; wait for its ready beep / steady LED.
- [ ] Power on the produce scale and complete the “PC” setup below if it is not already set.
- [ ] Open the browser and go to the OpenPantry **Scan (Laser)** page. It opens straight to the scanner — **no password** is needed, because the pantry network authorizes the station.
- [ ] Scan any item's barcode as a test — a “Last scan” line should appear. Then tap the red **× Cancel Order** so the test doesn't count.
- [ ] Tap once inside the barcode box so the cursor sits there (it glows green when focused).

Configuring the produce scale (one-time, each power-on)

The VEVOR scale with RS232 Port connects to the tablet as a USB keyboard. After you power it on it must be switched into **“PC” mode** so it types weights straight into OpenPantry. Do this every time the scale is powered up:

1	Power on the scale and let it settle to **0**.	Zero / stable reading
2	Press the **[.]** (period / decimal) key.	Enters setup
3	Wait until the display reads **“Entr”**.	Ready for unit
4	Press **[2]** to set the unit to **“lb”**.	Unit = lb
5	Press **[.]** (period), then press **[9]**.	Selects PC interface
6	Confirm the weight window shows **“PC”**.	Interfaced to OpenPantry

WHEN YOU SEE “PC”, YOU'RE SET

That reading means the scale is now talking to OpenPantry. Place a produce item on the scale during a scan and its weight fills the weight box automatically — no typing. If the window ever drops back to a plain weight, repeat the steps above.

Open-the-Station Checklist — Menu Counter

The Menu Counter has two screens: the **customer order form** that shoppers tap, and the **pick queue** that volunteers pull from in the back.

Customer-facing tablet

- [ ] Power on the customer tablet and connect it to the pantry Wi-Fi.
- [ ] Open the browser to the OpenPantry **Menu Counter** order form. It opens directly — **no password** to enter.
- [ ] Confirm the page loads the item buttons grouped by category (Dairy, Dry Goods, Frozen, …).
- [ ] Set the language selector back to English so the next shopper starts fresh.
- [ ] Stand the tablet in its holder facing the shopper; clean the screen.

Pick-queue tablet / screen (back of house)

- [ ] Open the **Menu Counter** → **Orders** (pick queue) page on the volunteer screen. It opens directly — **no password** needed.
- [ ] Confirm the queue auto-refreshes and the topbar shows “pending orders” and “items remaining”.
- [ ] Submit one test order from the customer tablet and confirm it appears in the queue, then Mark Complete to clear it.
- [ ] Make sure bags/bins and a pen are staged at the pick bench.

> **NO LOGIN ON THE PANTRY NETWORK**
> The Menu Counter order form and pick queue are secured by the pantry's network address, not a password, so volunteers never log in to use them. A password is only asked for on the administrator’s **item-setup** screen — leave that to the person opening the pantry.